Date:

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Q1 (a)

$$\begin{array}{r} 0110 \\ +\ 1100 \\ \hline 1010 \end{array} \qquad (1010)_2$$

Q2 (b)

$$\begin{array}{r} 11 \\ -01 \\ \hline 10 \end{array} \qquad (10)_2$$

Q3 (a) 0011 0010 1010 1001

hexadecimal

0011	0010	1010	1001
2^1+2^0	2^1	2^3+2^1	2^3+2^0
3	2	10	9
		A	

$= (32A9)_{16}$

Octal

011	001	010	101	001
3	1	2	5	1

$= (31251)_8$

Q4 (a) $(10A)_{16}$

$(10)_{10}$

$(0001)_2 + (0000)_2 + (1010)_2$

$= (100001010)_2$

2	10
2	5 – 0
2	2 – 1
	1 – 0

Date:

Q 5 (a) $(E5)_{16}$

(a) $14 \times 16^1 + 5 \times 16^0 = (229)_{10}$

(b) 229 -> Binary

1110 0101

011 100 101

3 4 5

2	229
2	114 - 1
2	57 - 0
2	28 - 1
2	14 - 0
2	7 - 0
2	3 - 1
2	1 - 1
	0

Q 6 (a) $(1234)_5$

converting to Base 10

$1 \times 5^3 + 2 \times 5^2 + 3 \times 5^1 + 4 \times 5^0$

$125 + 50 + 15 + 4 = (194)_{10}$

Converting to binary

1100 0010

12 2

C

$= (C2)_{16}$

2	194
2	97 - 0
2	48 - 1
2	12 - 0
2	6 - 0
2	3 - 0
2	1 - 1
	0

011 000 010

$3 + 0 + 2 = (302)_8$